

MIRROR PAGE

A comic-strip light-routing puzzle where an unreliable narrator twists the rules. • TGC Game Jam 2026 Proposal

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IndieConnect ID	[YOUR_INDIECONNECT_ID]	Public GitHub	github.com/SankeerthNara/TGC-Game-Jam (MIT)
Themes (All 3)	COMIC TWIST LIGHT	Target Playtime	12 – 15 mins first playthrough (20 min hard cap)
Engine & Renderer	Godot 4.7, GDScript only, GL Compatibility Renderer (Web/HTML5 export hosted on itch.io)		

1. PITCH & CORE CONCEPT

Every level is a physical page of a comic book. A light emitter casts a beam across panels to ignite target stars. The player solves each puzzle by **dragging comic panels to swap them** and **clicking mirrors to flip angles**. Because emitters, walls, mirrors, and targets reside inside individual panels, rearranging the layout fundamentally restructures the light's trajectory across gutters. Guiding the journey is a classic comic narrator who periodically breaks the fourth wall and rewrites reality.

2. INTEGRATION OF THE THREE THEMES

Theme	How It Drives Gameplay & Narrative
Comic	The page layout <i>is</i> the puzzle mechanic. Panels are movable physical tiles; gutters act as light refractors; captions deliver reactive narrative; sound effects appear as visual onomatopoeia (POW, ZAP, BAM); UI is styled after vintage newsprint comics.
Light	Light is the sole interactive agent and win condition. The beam reflects off angled mirrors, is blocked by inky walls, and activates star targets. Real-time path tracing provides immediate visual feedback.
Twist	Narrative & Mechanical Twists: (1) <i>Goal Twist (Level 3)</i> : Narrator reveals the hero you were lighting is the villain; stars and hazard targets swap roles, forcing an immediate re-solve. (2) <i>Rule Twists (Levels 4–6)</i> : Mirrors lie (reflecting backwards), gutters bend light 90°, and panels lock down.

3. CORE MECHANICS

- Panel Drag-Swap:** Click and drag any unlocked comic panel to swap positions with another. Everything inside the panel moves with it.
- Mirror Toggle:** Click rotatable mirrors (/ and \) to redirect the beam 90 degrees. Fixed mirrors (a, b) require panel relocation.
- Beam Tracing:** Headless raycast simulation that resolves instantaneously upon every move, lighting valid targets and alerting on hazards.
- Locked Panels:** Story-critical panels locked by narrator tape that cannot be moved, introducing spatial puzzle constraints.
- Frictionless Undo / Reset:** Instant [Z] undo and [R] reset encouraging playful experimentation without softlocks.

4. THE NARRATOR'S RULE TWISTS

Rule	Mechanical Impact	Introduced
Normal	Standard optical reflections: mirrors reflect as drawn; gutters maintain beam angle.	Levels 1, 2, 3
Goal Flip	Narrator caption flips polarity: good stars become toxic; hazards (red X) become the required target.	Levels 3 and 6
Lying Mirrors	Mirrors reflect the exact opposite of their drawn slant (slashes act as backslashes).	Level 4
Gutter Bend	Crossing a panel gutter forces a mandatory 90° clockwise beam deflection on the entry cell.	Levels 5 and 6

5. CONTENT PLAN & GAMEPLAY LOOP (15 MINUTES)

#	Level Name	Core Mechanic / Rule Introduced	Target Time
1	Page One	Panel swapping across gutters to carry a straight beam to the hero's star.	1.5 min
2	Looking Glass	Combining panel swapping with mirror angle toggles and wall obstacles.	2.5 min
3	The Twist	Goal Flip Twist: Solves normally, narrator flips roles, re-solve page for villain.	2.5 min
4	Liar, Liar	Lying Mirrors: Slashes reflect counter-intuitively; mental inversion puzzle.	2.5 min
5	Gutter Bend	Gutter Bend: Beams turn 90° at boundaries; locked panels limit swap freedom.	3.0 min
6	The Last Page	Grand Finale: Combines lying mirrors, gutter bends, locked tiles, and final flip.	3.0 min

Loop per Page: Read introductory narrator caption → Drag panels & flip mirrors → Light stars → Trigger reaction/twist → Turn page.

6. SCOPE BOUNDARIES

In Scope: 6 tightly tuned levels, panel swapping, mirror flipping, 3 rule mutations, goal flip twist, narrator caption UI, main menu, pause menu, end screen, retro comic sound effects, ambient music, web export.
Out of Scope (Strictly Cut): Level editor, save slots, mobile touch controls, gamepads, endless procedural mode.

7. TECHNICAL ARCHITECTURE & WEB FEASIBILITY

- **Engine & Renderer:** Godot 4.7 with GL Compatibility renderer; single-threaded Web export with zero native extensions or GDExtensions.
- **Model-View Decoupling:** Pure GDScript model (PageModel1, BeamSolver) isolated from rendering (PageView). Tested headless.
- **Automated Validation:** Built-in BFS/DFS solver (level_check.gd) verifies every level is 100% solvable within 2–8 optimal moves.
- **Web Export Optimization:** Web asset package is optimized to **under 460 KB PCK** (total uncompressed build 38.4 MB, ~8 MB gzip on itch.io). Verified with headless Playwright browser tests yielding 0 console errors.

8. 100-HOUR DEVELOPMENT MILESTONES

Hours	Phase & Deliverables	Current Status
0 – 12	Concept lock, core engine prototype, level format, art research, proposal submission.	COMPLETE
12 – 30	Vertical slice: Comic UI (HUD, caption box, pause), first art pass, CC0 audio, levels 1–4.	IN PROGRESS
30 – 60	Complete levels 5 & 6, gutter bend logic, locked panels, sound polish, music integration.	Planned
60 – 85	Balancing, narrative comedic timing, web playtests, UI juice (panel shake, onomatopoeia).	Planned
85 – 100	Feature freeze at Hour 85. Web export audit, itch.io upload, README and CREDITS audit.	Planned

9. ASSETS & AI DISCLOSURE

- **Asset Rule Compliance:** Zero paid assets. All assets are 100% free, CC0, or open source with full attribution in CREDITS.md:
 - **Fonts:** **Bangers** (SIL OFL 1.1) and **Comic Neue** (SIL OFL 1.1) from Google Fonts.
 - **Audio:** Procedural CC0 synthesized sound effects and jazz loop generated in Python.
 - **Art & Textures:** Vintage newsprint textures, Ben-Day halftone dots, and comic burst badges generated procedurally.
- **AI Collaboration:** All AI usage is documented in docs/AI_USAGE.md across Claude Code (architecture/core logic), Antigravity (UI, art/audio assets, browser QA), and ChatGPT (tests/docs). Core creative direction and design decisions remain human-authored.

10. RISKS & MITIGATIONS

Identified Risk	Proactive Mitigation Strategy
Web Export / Browser Bugs	Tested every milestone via Playwright headless Chromium. Compatibility renderer enforced from Day 1.
Puzzles Unsolvable / Obscure	Automated headless level validator verifies solvability and move counts before authoring.
Scope Overrun	Strict 6-level cap, 15-minute loop target, hard feature freeze at Hour 85.